PHASE 7

Power BI Data Model

From PostgreSQL analytics tables to a trusted interactive semantic model

B2B Travel Booking & Revenue Analytics

Complete Understanding Notes | Beginner-Friendly | Portfolio & Interview Ready

Purpose: understand what was done, why it was done, how it was validated, and how to explain it in an interview.

Prepared from the completed project work and verified analysis results.

Contents and Learning Map

Use this document in two ways: first as a learning summary, then as an interview revision sheet. Each section explains the concept, implementation, validation, and business meaning.

#	Section	What you should understand
1	Objective and loaded tables	Why modeling comes before visualization.
2	PostgreSQL connection and Power Query	Connection settings, data types, and common connection errors.
3	Date dimension	Why a dedicated date table is needed for time intelligence.
4	Relationships and grain	The final seven active relationships and filter direction.
5	Model validation	How slicers were used to prove relationships work.
6	Modeling principles and pitfalls	Many-to-many risk, auto-detection problems, and future date-model nuance.
7	Interview revision	How to explain the semantic model clearly.

Project rule: Do not memorize code alone. Be able to explain the grain of the data, why each step is needed, and how you verified the result.

1. Objective: Build the Semantic Model Before the Dashboard

Power BI is not only a charting tool. A reliable dashboard starts with a correct data model. Phase 7 connects to PostgreSQL, loads reporting tables, assigns correct data types, creates a date dimension, and defines relationships so filters propagate as intended.

Modeling principle: A beautiful visual built on the wrong relationship can still produce the wrong number. Model first, measure second, visualize third.

Tables loaded into Power BI

Table	Role	Grain
analytics.fact_bookings_fx	Primary booking fact	One valid row per booking
analytics.fact_cancellations_fx	Cancellation fact	One row per cancellation event
analytics.payment_booking_summary_fx	Payment summary	One row per booking
clean.dim_agents	Agent dimension	One row per agent
clean.dim_destinations	Destination dimension	One row per destination
clean.dim_hotels	Hotel dimension	One row per hotel
clean.dim_suppliers	Supplier dimension	One row per supplier

2. Connecting Power BI to PostgreSQL

The working connection used the local PostgreSQL server at 127.0.0.1:5432 and the database travel_analytics. Authentication issues were resolved before selecting the reporting tables in Navigator.

Issue encountered	Cause	Resolution
Authentication failed	Username typo: postgre instead of postgres	Use the correct PostgreSQL username.
No such host is known	Server/host value was not resolving correctly	Use 127.0.0.1:5432 for the local PostgreSQL instance.
Wrong data types after load	Power BI inferred some types automatically	Open Transform Data and explicitly set key/date/numeric types.

Important Power Query type checks

- booking_dt -> Date
- gross_booking_value_inr and other monetary metrics -> Decimal number
- booking_id, agent_id, hotel_id, supplier_id, destination_id -> Text
- rooms and nights -> Whole number where appropriate

3. Date Table and Time Intelligence Foundation

A dedicated date table is required for clean month sorting and DAX time-intelligence functions such as DATEADD. The date table covers the booking period from 1 Jan 2024 through 31 Dec 2025.

```
Date_Table =
ADDCOLUMNS(
    CALENDAR(DATE(2024,1,1), DATE(2025,12,31)),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month", FORMAT([Date], "MMM"),
    "Year Month", FORMAT([Date], "YYYY-MM"),
    "Quarter", "Q" & FORMAT([Date], "Q")
)
```

After creation, Date_Table was marked as the model date table. Month was sorted by Month Number to prevent alphabetical month ordering.

Why this matters: Without a proper date table, month sorting can be wrong and time-intelligence measures can behave unpredictably.

4. Final Relationship Model

Automatic relationship detection initially created incorrect links. Those were removed and the model was rebuilt deliberately. The final model contains exactly seven active relationships.

From (1 side)	To (* or 1 side)	Cardinality	Filter direction
dim_agents[agent_id]	fact_bookings_fx[agent_id]	1 : *	Single
dim_hotels[hotel_id]	fact_bookings_fx[hotel_id]	1 : *	Single
dim_suppliers[supplier_id]	fact_bookings_fx[supplier_id]	1 : *	Single
dim_destinations[destination_id]	fact_bookings_fx[destination_id]	1 : *	Single
Date_Table[Date]	fact_bookings_fx[booking_dt]	1 : *	Single

From (1 side)	To (* or 1 side)	Cardinality	Filter direction
fact_bookings_fx[booking_id]	payment_booking_summary_fx[booking_id]	1 : 1	Both (Power BI 1:1 behavior)
fact_bookings_fx[booking_id]	fact_cancellations_fx[booking_id]	1 : *	Single

Why these relationships are safe

Dimensions contain unique business keys and filter the booking fact. The payment summary was already aggregated to one row per booking before Power BI, so it can connect 1:1 without multiplying payment rows. Cancellation remains event-grain and therefore connects 1:* from booking to cancellation events.

Critical grain lesson: If raw payment transactions had been joined directly to bookings, bookings with multiple payments could have duplicated booking measures. Grain alignment was solved upstream in SQL.

5. Relationship Validation

A relationship should not be trusted only because a line appears in Model view. Interactive tests were used to verify filter propagation.

1. Add a Year slicer from Date_Table and a booking KPI card.
2. Select one year and confirm the booking card changes.
3. Add a destination slicer and confirm both booking and cancellation visuals respond.
4. Reset the slicers to All and verify the full totals return.

Test	Expected behavior	Result
Year slicer	Booking KPI changes by booking date	Validated
Destination slicer	Booking visuals filter to selected destination	Validated
Destination -> cancellations	Cancellation card changes through booking relationship	Validated

6. Modeling Principles, Pitfalls, and Future Improvements

Principle / pitfall	Explanation
Avoid incorrect auto relationships	Power BI may infer relationships between columns that share compatible data types but do not represent the same business entity.
Prefer single-direction dimension filters	Single direction makes the model easier to reason about and reduces ambiguity.
Use one row per dimension key	The 1-side must be unique for a clean star-schema relationship.
Keep facts at their natural grain	Bookings, payments, and cancellations represent different business events and should not be flattened carelessly.
Date range nuance	Date_Table ends Dec 2025 because it supports booking-date analysis. Cancellation events extend later; cancellation-date-specific analysis would need an extended/role-playing date model.
Unknown supplier handling	Missing supplier IDs were retained in the booking fact, so reports should label or handle unknown supplier values deliberately rather than silently dropping them.

7. Interview-Ready Model Explanation

60-second summary: I connected Power BI to the PostgreSQL analytics layer, loaded one primary booking fact, cancellation and payment facts, and four dimensions. I created a dedicated date table for time intelligence, removed incorrect auto-detected relationships, and built seven deliberate active relationships based on table grain and business keys. I validated the model using slicers to confirm filters propagated correctly. The key modeling decision was to aggregate payments to booking grain upstream so the Power BI model would not double-count booking metrics.

What you should be able to explain

- Difference between fact and dimension tables.
- Why booking_id can be 1:1 with payment_booking_summary_fx but 1:* with cancellation events.
- Why Date_Table is separate from the booking fact.
- Why auto-detected relationships can be dangerous.
- How you tested that the model works instead of assuming it works.

Phase 7 completion checklist

- Seven reporting tables loaded.
- Key data types checked.
- Date table created and marked.
- Month sorting configured.
- Exactly seven active relationships finalized.
- Incorrect auto relationships removed.
- Slicer-based QA completed.